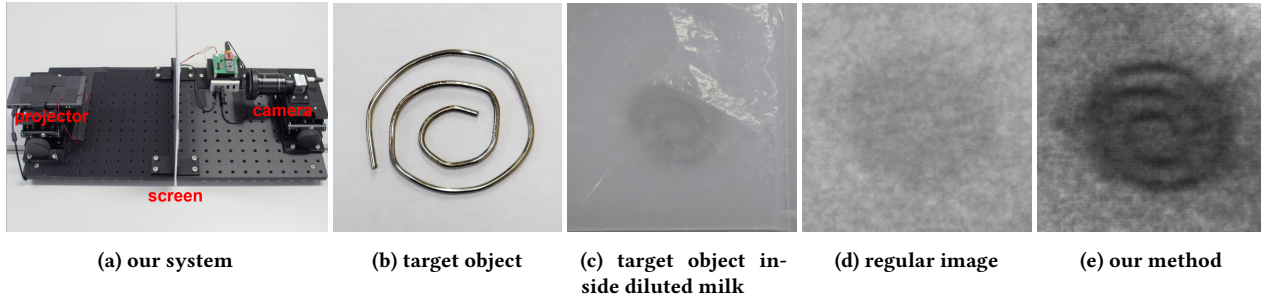


# Visibility Enhancement for Transmissive Image using Synchronized Side-by-side Projector–Camera System

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**Figure 1:** We built a side-by-side synchronized projector-camera system (a). We image a metal wire (b) in diluted milk (c). The visibility of the target object in a regular image (d) is significantly improved in the image obtained using our method(e).

## ABSTRACT

Extracting the direct light component from light transport helps to enhance the visibility of a scene. In this paper, we describe a method to improve the visibility of the target object by capturing transmissive rays without scattering rays using a synchronized projector–camera system. A rolling shutter camera and a laser raster scanning projector are placed side-by-side, and both epipolar planes are optically aligned on a screen plane which is place between the projector and camera. This paper demonstrates that our method can visualize an internal object inside diluted milk.

## CCS CONCEPTS

• Computing methodologies → Computational photography.

## KEYWORDS

datasets, neural networks, gaze detection, text tagging

### ACM Reference Format:

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## 1 INTRODUCTION

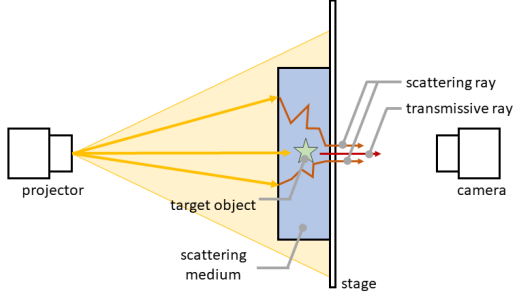
In a photograph taken at a heavily scattering scene such as in deep fog, a target object inside the scene can not be seen clearly because

of the scattering rays. O’Toole et al., [2015] developed a method to obtain an image of direct reflection component without global component such as scattering, to enhance the visibility of the target object using a synchronized projector-camera system which is called episcan3d. Moreover, Kubo et al., [2021] proposed a method to visualize under skin surface structure by capturing short range indirect component. Because the direction of the projector and the camera in both methods are same, they both focus on capturing reflection light component. However, to visualize a target object in a relatively thin scattering medium such as milky water, it is more reasonable to focus on the transmissible light component. Expanding previous researches, we built a specially designed optical setup to capture transmissive rays without scattering rays by synchronized projector camera system that the projector and the camera are placed side-by-side.

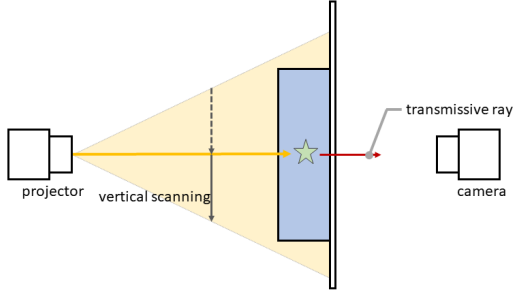
## 2 METHOD

Episcan3d[O’Toole et al. 2015] is a method which is possible to remove the scattered light component by synchronizing the timing of the projector’s raster scanning illumination and the rolling shutter camera’s read-out. We also use a raster scanning laser projector (SONY MP-CL1) and a rolling shutter camera (IDS UI-3240CP), both illumination and exposure timings are synchronized[O’Toole et al. 2015]. A schematic illustration of our method is show in Fig. 2. We placed the project and the camera side-by-side and both to be co-axial, and parallel. Because the optical alignment needs to be precise, we also prepared an special fixture using 3d printer for this purpose. Furthermore, we placed a transparent stage between them. The vertical scanning velocity is aligned on the stage by manipulating its pixel-clock manually. The measurement object is placed on the transparent stage, then the transmissive rays is captured. During the exposure of a horizontal camera sensor line, the projector illuminates only the corresponding horizontal line on

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(a) regular projector camera system



(b) our system

Figure 2: In regular system (a), because the project illuminates whole region in the field of view, the camera captures both transmissive ray and scattering rays. Instead, our synchronized projector-camera system (b) illuminates a single horizontal line at a time, the rolling shutter of camera can only captures the transmissive ray without scattering rays.

the stage. Therefore, most scattering rays cannot be observed in our system. Thus, the visibility of captured image is improved.

### 3 EXPERIMENTS

#### 3.1 metallic object inside diluted milk

To demonstrate the visibility enhancement by capturing transmissive rays, we image a metal wire (Fig. 1(b)) in a diluted milk (c) which is difficult to visualize in regular imaging(d). Instead, the visibility of the captured image of our method (e) is significantly improved. Figure 3 shows the pixel values on corresponding vertical lines of regular image and our method. It is obvious that the contrast of our method is improved.

#### 3.2 Thin leaves

We also image thin leaves using our optical system. Compared with the regular image shown in Fig. 4, the detail of the leaves are more visible by reducing the scattering rays inside.

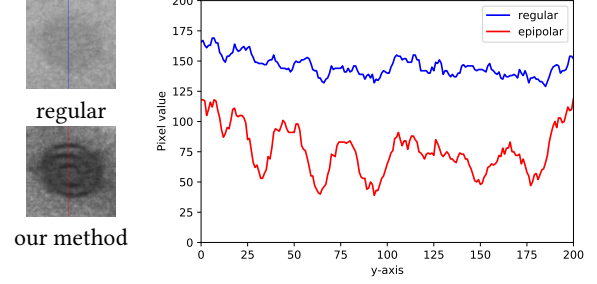
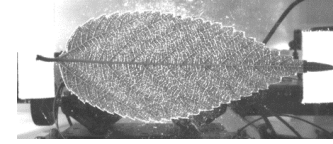
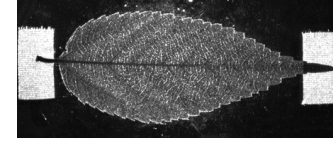


Figure 3: Evaluation of our imaging. We extract pixel values on blue and red lines from both regular image and our method shown on the left. The contrast of our method is significantly improved.



(a) regular photo



(b) our system

Figure 4: Imaging a thin leaf. Compared with a regular photo (a), fine details seem to be more visible in our method(b).

### 4 CONCLUSION

As a result of our experiments, we were able to eliminate scattered light and sharpen the image by using this method. This demonstrated that the Episcan can be used to capture images of scattered light without transmitted light.

### ACKNOWLEDGMENTS

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